

Calculus 3

Homework Assignment 1

Exercise 1. Find the limit, if it exists, or show that the limit does not exist.

$$(a) \lim_{(x,y) \rightarrow (1,2)} (5x^3 - x^2y^2)$$

$$(d) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2ye^y}{x^4 + 4y^2}$$

$$(b) \lim_{(x,y) \rightarrow (2,1)} \frac{4 - xy}{x^2 + 3y^2}$$

$$(e) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin^2 y}{x^2 + 2y^2}$$

$$(c) \lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - y^4}{x^2 + y^2}$$

$$(f) \lim_{(x,y,z) \rightarrow (0,0,0)} \frac{yz}{x^2 + 4y^2 + 9z^2}$$

Exercise 2. Find the first partial derivatives of the following functions.

$$(a) f(x, y) = y^5 - 3xy$$

$$(d) z = \tan(xy)$$

$$(b) f(x, t) = e^{-t} \cos(\pi x)$$

$$(e) f(x, y, z) = x \sin(y - z)$$

$$(c) f(x, t) = \sqrt{x} \ln t$$

$$(f) w = ze^{xyz}$$

Exercise 3. Find an equation of the tangent plane to each given surface at the specified point.

$$(a) z = 3y^2 - 2x^2 + x, \quad (2, -1, -3)$$

$$(c) z = \sqrt{xy}, \quad (1, 1, 1)$$

$$(b) z = 3(x - 1)^2 + 2(y + 3)^2 + 7, \quad (2, -2, 12)$$

$$(d) z = xe^{xy}, \quad (2, 0, 2)$$

Exercise 4. Explain why the function is differentiable at the given point, then find the linearization $L(x, y)$ at that point.

$$(a) f(x, y) = x^3y^4, \quad (1, 1)$$

$$(c) f(x, y) = \sqrt{x + e^{4y}}, \quad (3, 0)$$

$$(b) f(x, y) = \frac{x}{x + y}, \quad (2, 1)$$

$$(d) f(x, y) = e^{-xy} \cos y, \quad (\pi, 0)$$

Exercise 5. Use the Chain Rule to find dz/dt .

$$(a) z = x^2y, \quad x = t^2, \quad y = \sin t$$

$$(c) z = \sin(xy), \quad x = t^2, \quad y = \cos t$$

$$(b) z = \sqrt{x^2 + y^2}, \quad x = e^t, \quad y = \ln t$$

$$(d) z = e^{x+y}, \quad x = 1 + t, \quad y = 1 - t$$

Exercise 6. Use the Chain Rule to find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$.

$$(a) z = x^2y, \quad x = s + t, \quad y = s - t$$

$$(c) z = \ln(x^2 + y^2), \quad x = s + t, \quad y = s - t$$

$$(b) z = \arctan\left(\frac{x}{y}\right), \quad x = s^2 + t^2, \quad y = st$$

$$(d) z = \sqrt{x^2 + y^2}, \quad x = e^s \cos t, \quad y = e^s \sin t$$

Exercise 7. Use a tree diagram to write out the Chain Rule for the given case.

$$(a) \quad z = f(x, y), \quad x = g(u, v), \quad y = h(u, v) \quad h(u, v), \quad z = k(u, v)$$

$$(b) \quad w = f(x, y, z), \quad x = g(u, v), \quad y = (c) \quad z = f(x, y), \quad x = g(s, t), \quad y = h(s, t)$$

Exercise 8. (a) Let $p(t) = f(g(t), h(t))$ where f , g , and h are differentiable. Suppose $g(1) = 4$, $g'(1) = 2$, $h(1) = 5$, $h'(1) = -3$, $f_x(4, 5) = 6$, and $f_y(4, 5) = -2$. Find $p'(1)$.

(b) Let $R(s, t) = G(u(s, t), v(s, t))$, where G , u , and v are differentiable. Suppose

$$u(0, 1) = 3, \quad u_s(0, 1) = 2, \quad u_t(0, 1) = -4, \quad v(0, 1) = -2, \quad v_s(0, 1) = 1, \quad v_t(0, 1) = 5,$$

and

$$G_u(3, -2) = -1, \quad G_v(3, -2) = 4.$$

Find $R_s(0, 1)$ and $R_t(0, 1)$.

Exercise 9. Suppose f is a differentiable function of x and y . Define

$$g(r, s) = f(3r - s, s^2 - 2r).$$

Use the table of values below to compute $g_r(1, 1)$ and $g_s(1, 1)$.

(x, y)	f	f_x	f_y
$(2, -1)$	4	3	-2
$(5, 3)$	7	-1	5

Exercise 10. (a) If $z = f(x - y)$, show that

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0.$$

(b) If $z = f(x, y)$, where $x = s + t$ and $y = s - t$, show that

$$\left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2 = \frac{\partial z}{\partial s} \frac{\partial z}{\partial t}.$$

Exercise 11. Find the local maximum and minimum values and saddle point(s) of the function.

$$(a) \quad f(x, y) = y^3 + 3x^2y - 6x^2 - 6y^2 + 2$$

$$(c) \quad f(x, y) = x^3 - 12xy + 8y^3$$

$$(b) \quad f(x, y) = xy(1 - x - y)$$

$$(d) \quad f(x, y) = xy + \frac{1}{x} + \frac{1}{y}$$

Exercise 12. Use Lagrange multipliers to find the maximum and minimum values of the function subject to the given constraint.

$$(a) \quad f(x, y) = y^2 - x^2, \quad 4x^2 + y^2 = 1$$

$$(d) \quad f(x, y, z) = xyz, \quad x^2 + 2y^2 + 3z^2 = 6$$

$$(b) \quad f(x, y) = e^{xy}, \quad x^3 + y^3 = 16$$

$$(e) \quad f(x, y, z) = x^2y^2z^2, \quad x^2 + y^2 + z^2 = 1$$

$$(c) \quad f(x, y, z) = x^2 + y^2 + z^2, \quad x + y + z = 12 \quad (f) \quad f(x, y, z) = x^4 + y^4 + z^4, \quad x^2 + y^2 + z^2 = 1$$

Exercise 13. Find the absolute maximum and minimum values of f on the set D .

$$(a) \quad f(x, y) = 4x + 6y - x^2 - y^2, \quad D = \{(x, y) \mid 0 \leq x \leq 4, 0 \leq y \leq 5\}$$

$$(b) \quad f(x, y) = x^4 + y^4 - 4xy + 2, \quad D = \{(x, y) \mid 0 \leq x \leq 3, 0 \leq y \leq 2\}$$